

MRPL Refinery — Equipment Condition Monitoring Log: Centrifugal Pump P-204

Document Control

Document ID: MRPL-CM-P204-2026 | Classification: INTERNAL | Unit: Process Line A, Crude Distillation Unit | Reporting Period: Q3 2026

Equipment Specification

Parameter	Value
Equipment Tag	P-204
Equipment Type	Centrifugal Pump, Horizontal Split Case
Service	Crude Oil Feed Transfer
Location	Process Line A - CDU Unit 2
Manufacturer	KSB Process Pumps
Rated Speed	2960 RPM

Recent Vibration Readings

Vibration monitoring is performed weekly per SOP-017 using a handheld FFT analyzer on drive-end (DE) and non-drive-end (NDE) bearings. Alarm threshold: 7.1 mm/s RMS (ISO 10816-3, Zone C/D boundary for rigidly mounted medium machines). Warning band: 4.5-7.1 mm/s.

Date	DE Vibration (mm/s)	NDE Vibration (mm/s)	Bearing Temp (C)	Status
2026-08-25	3.8	3.5	68	Normal
2026-09-01	4.9	4.2	70	Warning
2026-09-08	6.3	5.1	72	Warning
2026-09-15	8.2	6.4	74	Critical

Latest Assessment (2026-09-15)

P-204 drive-end bearing vibration reached 8.2 mm/s RMS, exceeding the SOP-017 critical threshold of 7.1 mm/s. The increasing trend over three consecutive weekly readings (3.8 -> 4.9 -> 6.3 -> 8.2 mm/s) is consistent with progressive bearing race degradation. Bearing temperature has risen concurrently from 68C to 74C, corroborating the mechanical fault hypothesis. Per SOP-017 Section 3.2, any reading above 7.1 mm/s requires immediate isolation and engineer-approved shutdown before bearing failure risk escalates to seal damage or shaft scoring.

Recommended Action (per SOP-017 Section 3.2)

- Verify sensor calibration and confirm the vibration trend with a second reading.
- Isolate and shut down P-204 (requires HITL / Engineer approval per air-gapped safety protocol).

3. Perform mechanical inspection of bearing, coupling, and shaft alignment.
4. Replace drive-end bearing assembly if wear is confirmed; re-baseline vibration after reinstallation.